15 The graph of the curve C with equation $y = f(x)$ is transformed to give the graph of the curve S with equation $y = f(-x) - 3$

The point on C with coordinates (7, 2) is mapped to the point Q on S.

Find the coordinates of Q.

(..... , ...)

(Total for Question 15 is 2 marks)

16 Here are the first six terms of a quadratic sequence.

−1 5 15 29 47 69

Find an expression, in terms of n, for the nth term of this sequence.

First differences: 6, 10, 14, 18, 22.

1 mark
Constant second difference 4.

Second differences: 4, 4, 4, 4.

2a = 4, so a = 2; trial term $2n^2$.

1 mark
Leading term $2n^2$.

At n = 1, $2n^2$ = 2 but term = −1: subtract 3.

nth term = $2n^2 - 3$.

1 mark
$2n^2 - 3$.

$2n^2 - 3$

(Total for Question 16 is 3 marks)

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